

Aakash Thakkar

Software Engineer | Python, Platform & Developer Tooling

 Paderborn, Germany  p.aakash07@gmail.com  +49 16096235720  LinkedIn  GitHub

SUMMARY

Full stack engineer who builds and runs an internal platform used daily by around 25 engineers, in Python and TypeScript. FastAPI services and PostgreSQL and MongoDB behind them, React in the browser, Docker and Kubernetes underneath, with the monorepo, releases and GitHub Actions pipelines owned end to end. The platform configures operator software running on **60+ deployed industrial robot cells** at customers including WAGO, Albéa and Arvato. MSc Computer Science, Universität Paderborn.

EXPERIENCE

• **Full Stack Developer** 10.2022 – Present | Paderborn, Germany
Unchained Robotics GmbH 

Internal Platform and Developer Tooling

Python, FastAPI, TypeScript, React, Three.js, WebSockets, gRPC/Protobuf

- Main developer of a web-based configuration platform used daily by around 25 application engineers, whose output runs on 60+ deployed industrial cells in food, automotive and part-handling production (WAGO, Albéa, Arvato).
- Sets the technical direction for the platform layer, turning product requirements into the data model and the interaction contract those 25 engineers build against.
- Built a WebSocket-based locking mechanism that prevents two users editing the same project at once.
- Wrote Python simulators for hardware signals, so features can be developed, demonstrated and tested without access to physical hardware.
- Replaced ZMQ messaging with gRPC after recurring worker drops, defining the Protobuf schemas and the WebSocket bridge that streams events into the browser.
- Owns the pnpm monorepo and the release process: PR review, a major release every 3–4 months, minor releases roughly monthly.
- Runs sprint planning and capacity planning with the team lead, in a team of three.

Backend, Data and Infrastructure

FastAPI, PostgreSQL, MongoDB, Docker, Kubernetes, GitHub Actions, Scaleway, S3

- Wrote the FastAPI backend holding platform configuration, with PostgreSQL for runtime data and MongoDB for editor documents.
- Uses S3-compatible object storage for user-uploaded assets such as images and icons.
- Runs the cloud editor on managed Kubernetes at Scaleway, deployed through GitHub Actions, and debugs cluster and deployment problems as they come up.

Project Ownership: Sequencing and Geometry

TypeScript, React, computational geometry

- Led the rewrite of the automatic sequence generator from scoping through to release. The previous version produced sequences that collided, so operators had to verify every one by hand; the replacement uses a corridor-sweep method, runs client-side, and validates hand-written sequences at configurable clearance.
- Built a feasibility check that takes gripper and product dimensions, feeding orientation and the operator's selection, and works out whether items can be handled together in one movement or have to be split.

Hardware and Protocol Integration

Doosan, Universal Robots, Modbus TCP, OPC UA, Python

- Programs and commissions Doosan and Universal Robots cobots, scripting movements in Python over Modbus registers, against an in-house PLC platform speaking Modbus TCP and OPC UA.
- Second-level support for the controls engineers, tracing signal and communication faults across the interface, PLC and robot layers, including faults that turn out to sit elsewhere in the system.

• **Full Stack Developer (20h/week)** 03.2022 – 08.2022 | Paderborn, Germany
Unchained Robotics GmbH 

- Migrated an e-commerce platform from WordPress to NextJS, working with GraphQL against a Saleor backend.
- Added features to an existing web-based HMI, including the pallet schema editor, and worked on a Python updater thread pushing message-system updates to the database and HMI over WebSockets.

• **Full Stack Developer (20h/week)**

10.2020 – 12.2021 | Paderborn, Germany

Floccess GmbH

- Sole developer of a heat exchanger calculation web application (AngularJS, Django, SQL, Nginx), from build through testing to deployment.
- Added **ThreeJs** visualisations of exchanger vessels and interactive forms graphing calculated temperature and volume variations.

• **Student Research Assistant (20h/week)**

10.2019 – 09.2020 | Paderborn, Germany

Universität Paderborn

• **Systems Engineer**

2016 – 2019 | Delhi, India

Tata Consultancy Services Pvt. Ltd. 

- Built a Java tool called Legacy Adapter that pulls around **a million JSON messages a day** off a remote server, parses them and writes the extracted attributes into database tables, with automated tests flagging discrepancies against the source.
- Built mobile-compatible UI dashboards for retail data analysis in AngularJS and Node.js, and supported UNIX deployments with Git, SVN, SFTP and shell scripting.

EDUCATION

• **Master of Science - Computer Science**

2019 – 2024 | Paderborn, Germany

Universität Paderborn

- Master Thesis: Fuzzing with Static Analysis - Java, Maven, Bazel, Jazzer, JQF
Combined coverage-guided fuzzers with static data flow analysis, using the analysis to generate initial corpora and validate inputs.
- Linked Data to Natural Language (LD2NL) - Project Group
Built a summary generator for linked data (**Knowledge Graphs**) with a React frontend over a Java REST API, ordering output by node prominence.

• **Bachelor of Technology - Computer Science & Engineering**

2012 – 2016 | Vellore, India

VIT University

- Bachelor Project: Automatic Traffic Jam Detection
Wrote an image recognition program in Python and OpenCV that detects and quantifies traffic congestion in real time, then sets traffic signal states from current conditions.

LANGUAGES

English — C1 • **Deutsch** — B1

SKILLS

Robotics — Doosan and Universal Robots cobots, Modbus TCP, OPC UA, PLC step-chain logic, palletizing and depalletizing, machine vision (Inspekto)

Frontend — TypeScript, React, NextJS, Three.js, Tailwind

Backend — Python (FastAPI, Django), gRPC/Protobuf, WebSockets, REST, GraphQL

Languages — Python, TypeScript/JavaScript, Java

DevOps — Docker, Kubernetes, GitHub Actions, CI/CD, Git

Data — PostgreSQL, MongoDB

Process — Sprint and capacity planning, release management, PR review, Jira, Agile